CASE STUDY 6

ATM Transaction Simulation

Python Activity-Based Learning (ABL)

1. Problem Statement

Write a simple Python program to simulate an ATM. The user can check the balance, deposit money, or withdraw money. The program must not allow a withdrawal greater than the available balance or a deposit/withdrawal amount that is zero or negative.

2. Objectives

• Practise input and output statements.

• Use if/elif/else conditions to select a transaction.

• Update a balance after deposits and withdrawals.

• Apply simple validation to transaction amounts.

3. Assumptions

• The starting balance is ■5,000.

• The user chooses 1 to check balance, 2 to deposit, or 3 to withdraw.

• The user enters a whole-number amount for deposit or withdrawal.

• This is a classroom simulation; it does not connect to a real bank.

4. Algorithm

1. Start and set the balance to 5000.

2. Display the menu and read the user's choice.

3. If choice is 1, display the current balance.

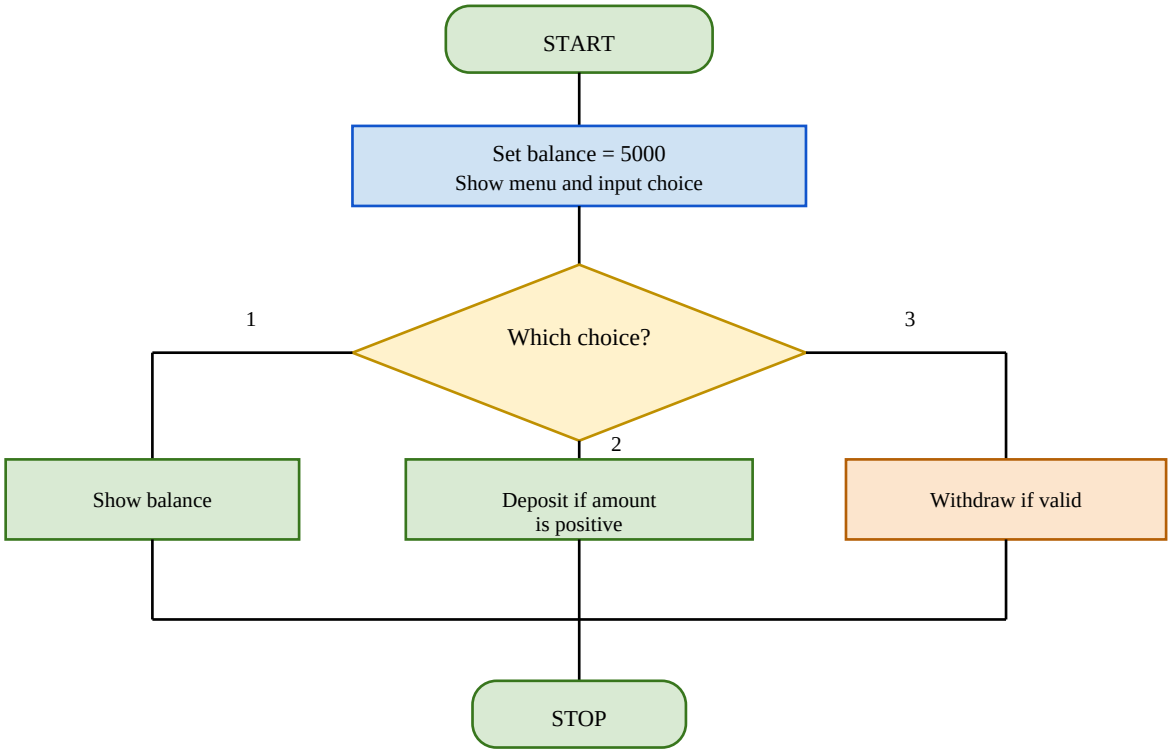
4. If choice is 2, read the deposit amount; add it if it is positive.

5. If choice is 3, read the withdrawal amount; withdraw only if it is positive and not more than the balance.

6. If the choice is not 1, 2, or 3, display an invalid-choice message.

7. Stop.

5. Colour Flowchart



6. Pseudocode

```
START
balance = 5000
INPUT choice

IF choice == 1
    PRINT balance
ELSE IF choice == 2
    INPUT amount
    IF amount > 0: balance = balance + amount
    ELSE: PRINT "Invalid amount"
ELSE IF choice == 3
    INPUT amount
    IF amount <= 0: PRINT "Invalid amount"
    ELSE IF amount > balance: PRINT "Insufficient balance"
    ELSE: balance = balance - amount
ELSE
    PRINT "Invalid choice"
STOP
```

7. Python Source Code

```
balance = 5000

print("1. Check Balance")
print("2. Deposit")
print("3. Withdraw")
choice = int(input("Enter your choice: "))

if choice == 1:
    print("Balance:", balance)
elif choice == 2:
    amount = int(input("Enter deposit amount: "))
    if amount > 0:
        balance = balance + amount
        print("New balance:", balance)
    else:
        print("Invalid amount")
elif choice == 3:
    amount = int(input("Enter withdrawal amount: "))
    if amount <= 0:
        print("Invalid amount")
    elif amount > balance:
        print("Insufficient balance")
```

```
    else:
        balance = balance - amount
        print("New balance:", balance)
else:
    print("Invalid choice")
```

8. Test Cases

Test	Choice / Input	Expected Output
1	1	Balance: 5000
2	2, amount 1000	New balance: 6000
3	3, amount 1500	New balance: 3500
4	3, amount 6000	Insufficient balance
5	2, amount 0	Invalid amount
6	7	Invalid choice

9. Sample Output (Output Screenshot Reference)

```
1. Check Balance
2. Deposit
3. Withdraw
Enter your choice: 2
Enter deposit amount: 1000
New balance: 6000
```

This is the expected sample output. For your submission, run the program in VS Code and take a screenshot of your own code and output.

10. Conclusion

This ATM simulation practises conditional statements and basic arithmetic. It checks transaction amounts and prevents withdrawals that are greater than the available balance.